

schtasks.exe

Cybersecurity Home Lab Guide

Windows Scheduled Tasks — Three Perspectives

ADMIN • DIGITAL FORENSICS • RED TEAM AWARENESS

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About This Lab

A short, practical walkthrough of schtasks.exe — a native Windows utility that matters to every cybersecurity role. This lab documents the same tool through three distinct lenses: system administration, digital forensics, and red team awareness. The goal is not to memorise commands, but to develop the right mindset for each context.

OVERVIEW

What is schtasks?

Core utility — six verbs, unlimited context

schtasks.exe is a native Windows command-line utility for creating, querying, modifying, and deleting scheduled tasks. It interacts directly with the Windows Task Scheduler service and is an essential tool for system administrators, digital forensic analysts, and penetration testers alike.

Command Verb	What It Does
<code>/Create</code>	Register a new scheduled task on the system
<code>/Query</code>	List and inspect existing scheduled tasks
<code>/Change</code>	Modify the properties of an existing task
<code>/Run</code>	Manually trigger a task immediately, outside its schedule
<code>/End</code>	Stop a currently running task instance
<code>/Delete</code>	Remove a task from the Task Scheduler permanently

Orientation

Privilege: Standard User

Before touching anything, we establish a baseline. Running `schtasks /?` displays all available flags, verbs, and syntax examples — this is our reference card for everything that follows.

```
C:\Windows\System32>schtasks /?  
  
SCHTASKS /parameter [arguments]  
  
Description:  
    Enables an administrator to create, delete, query, change, run and  
    end scheduled tasks on a local or remote system.  
  
Parameter List:  
    /Create          Creates a new scheduled task.  
    /Delete          Deletes the scheduled task(s).  
    /Query           Displays all scheduled tasks.  
    /Change          Changes the properties of scheduled task.  
    /Run            Runs the scheduled task on demand.  
    /End            Stops the currently running scheduled task.  
    /ShowSid        Shows the security identifier corresponding to a scheduled task name.  
    /?             Displays this help message.  
  
Examples:  
    SHTASKS  
    SHTASKS /?  
    SHTASKS /Run /?  
    SHTASKS /End /?  
    SHTASKS /Create /?  
    SHTASKS /Delete /?  
    SHTASKS /Query /?  
    SHTASKS /Change /?  
    SHTASKS /ShowSid /?
```

As shown in the output above, we can specify the action and also observe, what flags are compatible with that

```
C:\Windows\System32>schtasks /create /?
```

```
SCHTASKS /Create [/S system [/U username [/P [password]]]]
[/RU username [/RP password]] /SC schedule [/MO modifier] [/D day]
[/M months] [/I idletime] /TN taskname /TR taskrun [/ST starttime]
[/RI interval] [ {/ET endtime | /DU duration} [/K] [/XML xmlfile] [/V1]]
[/SD startdate] [/ED enddate] [/IT | /NP] [/Z] [/F] [/HRESULT] [/?]
```

Description:

Enables an administrator to create scheduled tasks on a local or remote system.

Parameter List:

/S system	Specifies the remote system to connect to. If omitted the system parameter defaults to the local system.
/U username	Specifies the user context under which SchTasks.exe should execute.
/P [password]	Specifies the password for the given user context. Prompts for input if omitted.
/RU username	Specifies the "run as" user account (user context) under which the task runs. For the system account, valid values are "", "NT AUTHORITY\SYSTEM" or "SYSTEM". For v2 tasks, "NT AUTHORITY\LOCALSERVICE" and "NT AUTHORITY\NETWORKSERVICE" are also available as well as the well known SIDs for all three.

<SNIP>

Examples:

==> Creates a scheduled task "doc" on the remote machine "ABC" which runs notepad.exe every hour under user "runasuser". <SNIP>

```
SCHTASKS /Create /S ABC /U user /P password /RU runasuser
/ RP runaspassword /SC HOURLY /TN doc /TR notepad
```

==> Creates a scheduled task "accountant" on the remote machine "ABC" to run calc.exe every five minutes from the specified start time to end time between the start date and end date.

```
SCHTASKS /Create /S ABC /U domain\user /P password /SC MINUTE
/ MO 5 /TN accountant /TR calc.exe /ST 12:00 /ET 14:00
/ SD 06/06/2006 /ED 06/06/2006 /RU runasuser /RP userpassword
```

==> Creates a scheduled task "gametime" to run freecell on the first Sunday of every month.

```
SCHTASKS /Create /SC MONTHLY /MO first /D SUN /TN gametime
/ TR c:\windows\system32\freecell
```

<SNIP>

Admin Perspective

Privilege: Run as Administrator

Goal: Create, inspect, modify, and delete scheduled tasks the way a Windows sysadmin would. Automation reduces manual workload and minimises human error on routine maintenance tasks.

► Step 1 — Create a Task

We create an automated Disk Cleanup task to run daily. Less manual intervention, more time for higher-priority work.

```
C:\Windows\System32>schtasks /create /tn "LabTest\DailyCleanUp" /tr "C:\Windows\System32\cleanmgr.exe" /sc daily /st 15:55 /ru system
SUCCESS: The scheduled task "LabTest\DailyCleanUp" has successfully been created.
```

Flag breakdown:

Flag	Meaning
/TN	Task Name — use a folder\name structure to organise tasks (e.g. LabTest\TaskName)
/TR	Task Run — the executable, script, or command to execute
/SC	Schedule type: MINUTE, HOURLY, DAILY, WEEKLY, MONTHLY, ONCE, ONSTART, ONLOGON, ONIDLE, ONEVENT
/ST	Start Time in 24-hour format HH:MM
/RU	Run as User — SYSTEM for elevated execution, or specify a named account

► Step 2 — Verify the Task

Use /Query to confirm the task was registered correctly. The /FO LIST flag formats the output as a readable list rather than a wide table.

```
C:\Windows\System32>schtasks /query /tn "LabTest\DailyCleanUp" /fo list

Folder: LabTest
HostName:      WINDOWS-11SH
TaskName:      LabTest\DailyCleanUp
Next Run Time: 6/6/2026 3:55:00 PM
Status:        Ready
Logon Mode:    Interactive/Background
```

Adding /V (verbose) expands the output to include trigger details, last run time, next run time, task status, and the run-as account — the most complete view available from the command line.

```
C:\Windows\System32>schtasks /query /tn "LabTest\DailyCleanUp" /fo list /v

Folder: LabTest
HostName: WINDOWS-11SH
TaskName: LabTest\DailyCleanUp
Next Run Time: 6/7/2026 3:55:00 PM
Status: Running
Logon Mode: Interactive/Background
Last Run Time: 6/6/2026 3:55:00 PM
Last Result: 267009
Author: WINDOWS-11SH\vboxuserw
Task To Run: C:\Windows\System32\cleanmgr.exe
Start In: N/A
Comment: N/A
Scheduled Task State: Enabled
Idle Time: Disabled
Power Management: Stop On Battery Mode, No Start On Batteries
Run As User: SYSTEM
Delete Task If Not Rescheduled: Disabled
Stop Task If Runs X Hours and X Mins: 72:00:00
Schedule: Scheduling data is not available in this format.
Schedule Type: Daily
Start Time: 3:55:00 PM
Start Date: 6/6/2026
End Date: N/A
Days: Every 1 day(s)
Months: N/A
Repeat: Every: Disabled
Repeat: Until: Time: Disabled
Repeat: Until: Duration: Disabled
Repeat: Stop If Still Running: Disabled
```

► Step 3 — Modify the Task

To change the start time without recreating the task, use `/Change` with the specific parameter to update. Here we shift the Disk Cleanup window from 15:55 to 16:05.

```
C:\Windows\System32>schtasks /change /tn "LabTest\DailyCleanUp" /st 16:05
SUCCESS: The parameters of scheduled task "LabTest\DailyCleanUp" have been changed.
```

Always re-run the query after a change to confirm it was applied correctly. Verification is not optional — it is the discipline that separates rigorous operators from careless ones.

```
C:\Windows\System32>schtasks /query /tn "LabTest\DailyCleanUp" /fo list

Folder: LabTest
HostName: WINDOWS-11SH
TaskName: LabTest\DailyCleanUp
Next Run Time: 6/6/2026 4:05:00 PM
Status: Running
Logon Mode: Interactive/Background
```

► Step 4 — Trigger Manually

If a task needs to run immediately — outside its scheduled window — `/Run` forces execution on demand. Useful for testing and incident response scenarios.

```
C:\Windows\System32>schtasks /run /tn "LabTest\DailyCleanUp"
INFO: scheduled task "LabTest\DailyCleanUp" is currently running.
SUCCESS: Attempted to run the scheduled task "LabTest\DailyCleanUp".
C:\Windows\System32>
```

► Step 5 — Delete the Task and Verify

When a task is no longer needed, /Delete removes it permanently. The /F flag suppresses the confirmation prompt — useful for scripting, but should be handled with care in production.

```
C:\Windows\System32>schtasks /delete /tn "LabTest\DailyCleanUp" /f
SUCCESS: The scheduled task "LabTest\DailyCleanUp" was successfully deleted.

C:\Windows\System32>schtasks /query /tn "LabTest\DailyCleanUp"
ERROR: The system cannot find the path specified.
```

Digital Forensics Perspective

Duration: ~20 min — Privilege: Run as Administrator

Goal: Enumerate all scheduled tasks system-wide, export them as forensic evidence, and identify indicators of suspicious scheduling activity. In a real investigation, this process forms part of the artefact collection phase.

► Step 1 — Export All Tasks to CSV

A full verbose export captures every scheduled task with complete detail. The resulting file becomes a structured evidence artefact that can be reviewed in Excel, ingested into a SIEM, or attached to an incident report.

```
C:\Windows\System32>schtasks /query /fo csv /v
> C:\Users\%Username%\Desktop\all_tasks.csv
```

► Step 2 — Filter Task Executables

findstr /i filters the verbose output to show only lines matching a keyword (-s). This is the single most forensically relevant field in a scheduled task: it tells us **information of interest**.

```
C:\Windows\System32>schtasks /Query /FO LIST /V | findstr /i "Task To Run"
TaskName: \MicrosoftEdgeUpdateTaskMachineCore{8891E65B-8FA9-4813-883F-93058E3D99FF}
Next Run Time: 6/6/2026 11:19:56 PM
Status: Running
Last Run Time: 6/6/2026 3:29:06 PM
Task To Run: C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe /c
Comment: Keeps your Microsoft software up to date. If this task is disabled or stop
l not be kept up to date, meaning security vulnerabilities that may arise cannot be fixed and features may not w
wh
Scheduled Task State: Enabled
Power Management: Stop On Battery Mode
Run As User: SYSTEM
Delete Task If Not Rescheduled: Disabled
Stop Task If Runs X Hours and X Mins: 72:00:00
Repeat: Stop If Still Running: N/A
```

Note: findstr /i performs a case-insensitive match. We can use different regular expressions (regex) to further narrow the scope of our search.

Legitimate tasks run from System32 or Program Files. The following patterns warrant immediate investigation:

- %APPDATA% or %TEMP% directories
- .bat, .vbs, or .ps1 script files with non-descriptive names
- Double-extension filenames such as update.exe.bat
- Paths inside C:\Users\Public or other user-writable directories

Any task running as SYSTEM or a privileged account that does not correspond to a known application deserves scrutiny.

► Cross-Reference: Windows Event Logs

Scheduled task activity is logged under:

```
Event Viewer → Applications and Services Logs →  
Microsoft → Windows → TaskScheduler → Operational
```

Or query directly from the command line:

```
wevtutil ql Microsoft-Windows-TaskScheduler/Operational /f:text /c:20
```

Key Event IDs every analyst should know:

Event ID	Meaning
106	Task Registered — a new scheduled task was created on the system
200	Task Action Started — a task began executing
201	Task Action Completed — a task finished its run
141	Task Deleted — a task was removed from the scheduler
102	Task Instance Completed — task run succeeded

Key Forensic Insight

Attackers frequently delete tasks after establishing persistence to cover their tracks. Event ID 141 records the deletion event — meaning even after the task is gone, the event log proves it existed. Always export the TaskScheduler Operational log as part of our evidence collection.

Red Team Awareness

Isolated Lab VM Only

Goal: Simulate how an attacker might use schtasks for persistence, so we can build detection rules and recognise anomalies in real environments.

MITRE ATT&CK® T1053.005 — Scheduled Task/Job

► Technique 1 — Basic Persistence Simulation

An attacker who has already gained code execution on a target machine may plant a scheduled task that fires on every logon, ensuring persistence across reboots and user sessions.

```
C:\Windows\System32>schtasks /create /tn "WindowsUpdate\Updater" /tr "powershell.exe -WindowsStyle Hidden -Command Write-Host 'simulated'" /SC ONLOGON /RU SYSTEM
SUCCESS: The scheduled task "WindowsUpdate\Updater" has successfully been created.
```

Red flags to recognise:

Indicator	Why It Is Suspicious
WindowsUpdate\Updater	Masquerades as a legitimate Windows component — a classic Living-off-the-Land naming convention
-WindowStyle Hidden	Suppresses the PowerShell window, hiding execution from the logged-in user
/SC ONLOGON	Triggers on every user login — guarantees persistence across reboots
/RU SYSTEM	Runs with SYSTEM-level privileges, maximising potential impact

None of these indicators are automatically malicious in isolation — but their combination, especially in an unexpected context, absolutely warrants investigation.

► Technique 2 — Remote Task Creation (Localhost Only)

The /S flag allows schtasks to target a remote machine. After compromising credentials, an attacker can register tasks on other hosts in the network without ever touching them physically — a lateral movement technique.

```
C:\Windows\System32>schtasks /Create /TN "LabTest\RemoteTest" /TR "calc.exe" /SC ONCE /ST 00:00 /S localhost /U %USERNAME% /P *
```

In this lab, /S localhost keeps execution contained and safe. In a real attack, localhost would be replaced with a remote hostname or IP address.

► Detection Checklist

We should use the following checklist when threat-hunting for malicious scheduled tasks:

- Tasks created from %TEMP%, %APPDATA%, or C:\Users\Public
- PowerShell, wscript, or cscript listed as the task action executable
- Tasks with no Description or Author metadata field populated
- ONLOGON or ONSTART triggers on tasks created by non-system processes
- Task names closely resembling known Windows folders (WindowsUpdate\, Microsoft\)
- Event ID 141 (deletion) immediately following Event ID 106 (creation)

CONCLUSION

Summary & Key Takeaways

What this lab teaches beyond the syntax

schtasks.exe is a single binary that ships with every Windows installation. What makes it interesting — and worth studying early in cybersecurity journey — is not its syntax. It is how differently the same tool looks depending on where we are standing.

What Each Perspective Taught Me

Admin

Scheduled tasks are infrastructure. Creating, modifying, and deleting them with discipline — using folder namespaces, verifying changes, and understanding /RU contexts — is fundamental Windows administration. A sysadmin who does not audit their own scheduled tasks is leaving a blind spot that defenders and attackers will both eventually find.

Digital Forensics

The forensic value of schtasks lies in what it reveals about a system's intended behaviour over time. Bulk CSV exports, XML artefacts, and Event Log correlation (especially Event IDs 106 and 141) paint a clear picture of what was scheduled, when, and by whom — even after a task has been deleted. Every artefact tells a story; the analyst's job is to read it.

Red Team Awareness

T1053.005 is one of the most consistently abused persistence techniques in real-world intrusions. Its effectiveness comes from legitimacy — schtasks is a trusted Windows binary, and a well-named task blends into the noise of hundreds of legitimate ones. Detection requires both behavioral awareness (what triggers look suspicious?) and artefact knowledge (what did the Event Log record?). We cannot defend what we do not understand.

The Core Lesson

Cybersecurity knowledge is not siloed. The same command we use to create a Disk Cleanup task is almost the same one an attacker uses to maintain access after a breach. The same query that helps a sysadmin verify their automation is the same query a forensic analyst runs during triage. Learn the tool once — understand it from every angle.

The End.